MILITARY SPECIFICATION

FUZE, MINE, COMBINATION M6A1 AND M10A1, PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of Fuze, Mine, Combination designated M6A1 and M10A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-P-10025 — Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

73-9-13 — Fuze, Mine, Combination M6A1 Assembly and Details.

73-9-25 — Fuze, Mine, Combination M10A1 Assembly and Details.

PUBLICATIONS

ORDNANCE CORPS

IEL-73-9-25A — Index of Inspection Equipment Lists, M10A1 Fuze.

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IEL–73–9–13A — Index of Inspection Equipment Lists, M6A1 Fuze.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The parts assembly shall comply with all requirements specified on Drawings (dwg.) 73–9–13 and 73–9–25 and with all requirements specified in applicable specifications.

3.3 Functioning, release pin. The fuze assembly, less igniter assembly shall function when a maximum pull force of 10 pounds is applied to the release pin ring, and shall not function when a minimum pull force of three pounds is applied to the release pin ring when tested as specified in 4.4.1.

3.4 Functioning, prongs. The fuze assembly less igniter assembly shall function when a maximum load of 20 pounds is applied to the prongs, and shall not function when a minimum load of eight pounds is applied to the prongs when tested as specified in 4.4.2.

3.5 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analyses on all material purchased directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be sub-

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stantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector will verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to normal production the contractor shall submit an initial production sample consisting of (fifty) assemblies and complete sets of parts (i.e. 50 each of every individual component and every subassembly), which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of analyses for all materials. A sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Each assembly and component shall be inspected by the Government for all the inspections specified in 4.3.2 and 4.3.3 of this specification and for all additional requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the contracting officer.

4.3 *Inspection provisions.*

4.3.1 *Lot formation.* A lot shall consist of parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Base (see dwg. 73–9–14, covering a detail of dwg. 73–9–13 or dwg. 73–9–25).

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Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Vent holes missing or obstructed (73–9–25D only)	Visual	01001
Major:	AQL 1.50 percent		
101.	Diameter of flash hole, minimum (min.)	Gage	01002
102.	Pitch diameter of large thread, maximum (max.)	Gage	01003
103.	Major diameter of large thread, max	Gage	01004
104.	Pitch diameter of small thread, max	Gage	01005
105.	Major diameter of small thread, max	Gage	01006
106.	Diameter of primer cavity, min	Gage	01007
107.	Depth of primer cavity, min	Gage	01008
108.	Outside diameter of igniter tube end, max	Gage	01009
109.	Diameter of safety fuze cavity, min. (73–9–25D only)	Gage	01010
110.	Concentricity between igniter tube end and pitch diameter of large thread	Gage	01011
111.	Profile of primer cavity (Dwg. 73–9–14D only)	Gage	01012
Minor:	AQL 4.00 percent		
201.	Total length, max	Gage	01013
202.	Length from igniter tube end to shoulder under large thread, max. (dwg. 73–9–25D only)	Gage	01014
203.	Diameter of flange, min	Gage	01015
204.	Diameter of crimping groove (dwg. 73–9–14D only)	Gage	01016
205.	Location of crimping groove (73–9–14D only)	Gage	01017
206.	Foreign matter	Visual	01018
207.	Finish improper	Visual	01019
208.	Protective coating inadequate	Visual	01020

4.3.2.2 Fuze assembly (less igniter assembly) (see dwg. 73–9–13 or dwg. 73–9–25).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Safety pin missing or incorrectly positioned	Visual	02001
Major:	AQL 0.65 percent		
101.	Pitch diameter of internal thread, min	Gage	02002
102.	Minor diameter of internal thread, min	Gage	02003
103.	Crimp of case to head inadequate	Visual	02004
104.	Any prong missing or insecure	Visual	02005
105.	Crimp of head over trigger pin inadequate	Visual-Manual	02006
Minor:	AQL 1.50 percent		
201.	Head not free to turn after crimping to case	Manual	02007
202.	Protective coating inadequate	Visual	02008
203.	Foreign matter	Visual	02009

4.3.3 Testing.

4.3.3.1 *Functioning, release pin, critical defect if 02010 assembly functions at minimum pull, major defect if assembly does not function at maximum pull or less.* This test shall be performed 100 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4 (see 6.4).

4.3.3.2 *Functioning, prongs, critical defect if assembly 02011 functions at minimum load, major defect if assembly does not function at maximum load or less.* This test shall be performed 100 percent. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4 (see 6.4).

4.3.4 Inspection Equipment. Index of Inspection Equipment Lists Number IEL–73–

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9–25A and IEL–73–9–13A identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance. Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 Government use of contractor's inspection equipment. The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.4 Test methods and procedures.

4.4.1 *Functioning, release pin.* The fuze assembly, less igniter assembly, shall be subjected to a pull of three pounds applied at a rate of one foot minimum per minute to the release pin ring parallel to the axis of the release pin. Any assembly that functions does not comply with 3.3 and shall be removed from the lot. The fuze assembly, less igniter assembly, shall be subjected to a pull of 10 pounds applied at a rate of one foot maximum per minute to the release pin ring parallel to the axis of the release pin. Any assembly that fails to function does not comply with 3.3 and shall be removed from the lot.

4.4.2 *Functioning, prongs.* The fuze assembly, less igniter assembly, shall be subjected to a load of eight pounds applied at a rate of one foot minimum per minute to any one prong or equally distributed to all three prongs parallel to the axis of the assembly. Any assembly that functions does not comply with 3.4 and shall be removed from the lot. The fuze assembly, less igniter assembly, shall be subjected to a load of 20 pounds applied at a rate of one foot maximum per minute to any one prong, or equally distributed to all three prongs parallel to the axis of the assembly. Any assembly that fails to function does not comply with 3.4 and shall be removed from the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C. The parts and assemblies for Fuze, Mine, Combination M6A1 and M10A1 shall be packaged in accordance with Specification MIL–P–10025 in such a manner that no damage will occur during shipment.

5.2 Packing.

5.2.1 *Level C.* The parts and assemblies for Fuze, Mine, Combination M6A1 and M10A1 shall be packed in accordance with Specification MIL–P–10025. Separately designated items shall not be packed together.

5.3 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

5.4 Shipping. When component parts from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated.

5.5 Marking and labeling. Marking of cartons shall be in accordance with Standard MIL–P–10025.

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6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor will design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection Equipment Lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection Equipment List codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM–608–12.

CDCF — Contractor's design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has

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universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists will be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Functioning. The fuze assembly, less igniter assembly, shall be considered to have functioned if the firing pin is released and capable of clearing the assembly when tested.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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